CardiomyoML: Classification of cardiomyopathies and prediction of new biomarkers with machine learning using RNA sequencing data

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Fellowship of the MAGNet:
Sebastián Ayala[1] - s.ayalaruano@student.maastrichtuniversity.nl
Sonia Bălan[1] - s.balan@student.maastrichtuniversity.nl
Diego Rodríguez[1] - d.rodriguezesperante@student.maastrichtuniversity.nl
Rodrigo Sánchez[1] - rodrigo.sanchez@student.maastrichtuniversity.nl

[1] *Maastricht University - MSc Systems Biology*

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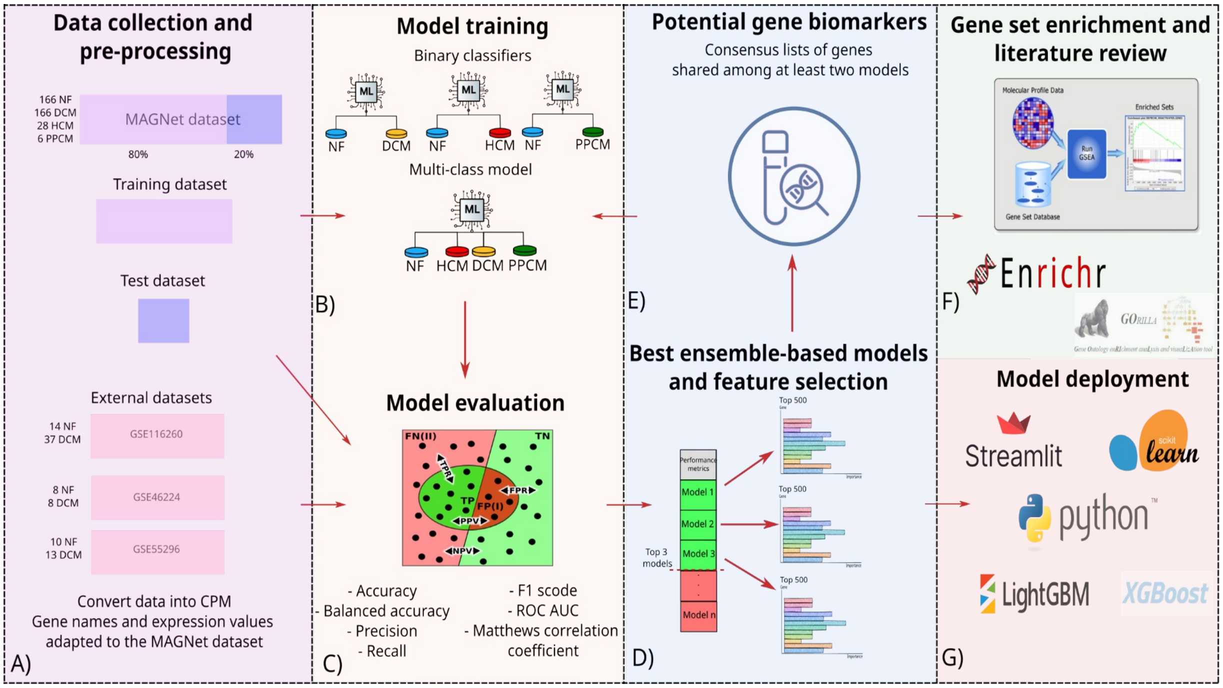
Abstract

Cardiomyopathies affect millions of people worldwide. Although there is a clear genetic component to cardiomyopathies, the underlying effects on the subtypes of these conditions are still not completely understood. In this study we created machine learning (ML) models trained with gene expression data with the aim to classify patients according to their etiology, distinguishing between non-failing (NF) and either dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM) or peripartum cardiomyopathy (PPCM). We aimed to find relevant genes through feature selection and interpreted the biological relevance of these features via gene set enrichment and a literature review of the most important genes. The models performed extremely well in the binary classification of NF/DCM, and still performed well in the binary classifications of NF/HCM and NF/PPCM using the test data. The multiclass classifier seemed to perform well but was in fact not able to distinguish the HCM and PPCM cases. However, the models did not perform better than a random classifier when using external validation data. Several improvements need to be made to our current method (e.g., alternative normalization, batch correction strategies for data preprocessing, among others), and standardization of data handling methods would be beneficial before this model can truly be employed in a clinical setting. Nonetheless, such ML approaches have great potential for diagnosis and the identification of potential biomarkers, drug targets, and genes of interest. To guarantee the reproducibility and accessibility of our pipeline, we launched a web application with the best binary classifier for NF/DCM.



Introduction

Heart failure is a heterogeneous disease affecting tens of millions of people worldwide[1]. One of the factors that can cause heart failure is cardiomyopathies, which are morphological and functional abnormalities in the myocardium in the absence of other diseases[2,3]. Classification within cardiomyopathies is disputed, but the most common types reported in clinical settings are dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and restricted cardiomyopathy[2–4]. There are also several subtypes and unclassified cardiomyopathies like peripartum cardiomyopathy (PPCM) which is a subtype of DCM that can occur due to the stress of pregnancy[2,3]. Cardiomyopathies can then be divided into familial or non-familial[2,4]. The genetic dimension of cardiomyopathies has become more relevant throughout the years, to the point that some experts recommend classifying cardiomyopathies entirely based on the affected genes and encoded proteins[3]. Although this classification might not be the most efficient for diagnosis in a clinical setting, it is undeniable that genetics heavily influence the phenotype of cardiomyopathies[3].

In recent years, machine learning (ML) algorithms have been used in biomedical research to classify and investigate many different diseases using RNA sequencing

(RNA-seq) data[5–11], including a study that employed ML models to classify non-failure and DCM samples[12]. Not only have ML algorithms proved to be useful tools in analyzing RNA-seq and transcriptomics data in general, but ML feature selection has been shown to outperform classical differential gene expression analyses (DGEA) in predicting which genes are relevant to a given phenotype as they consider the relationships between genes instead of treating them as independent[13]. Ensemble learning is a ML strategy that combines the predictions of multiple models to improve the performance of individual estimators[14,15].

This report presents an analysis of the capabilities of ML algorithms to distinguish between failing and non-failing (NF) heart tissue and extract genes of interest from RNA-seq data. This study can be divided into five steps. First, gene expression datasets were extracted and processed where appropriate. Second, ML-based binary and multiclass classifiers were trained and evaluated using various performance metrics. Third, a consensus list of the most important genes from the top 3 ensemble-based models was obtained, and a new model was trained to test the predictive power of these genes. Fourth, the list of most important genes was analyzed via gene set enrichment (GSE) and was compared to the existing literature for relevant links to cardiomyopathies. Finally, a web application with the best binary classifier for DCM was created.

Methods

Datasets

The Myocardial Applied Genomics Network (MAGNet) dataset was used to train the ML algorithms. This dataset (GSE141910) includes 366 subjects, of whom 166 were non-failing (NF), 166 were diagnosed with DCM, 28 with HCM, and 6 with PPCM[16]. The samples came from left ventricular cardiac tissue, the NF samples were collected from unused donor hearts and the failing samples were taken from patients undergoing open heart surgery. RNA was collected from these samples and sequences were obtained using Illumina 2500. The data was presented in counts per million (CPM).

External validation datasets were obtained from the Gene Expression Omnibus (GEO) and ArrayExpress. The search terms used were “RNA-seq” and "cardiomyopathy". Further filtering steps involved looking for datasets whose samples were taken from left ventricular heart tissue using an RNA-seq technique, in order to match the MAGNet dataset. Three datasets were obtained this way: GSE46224[17,18], GSE116250[19,20], and GSE55296[21,22]. All these datasets deal mostly with NF and DCM samples. Information regarding the number of samples in each etiology, sequencing platform, and RNA expression measure unit can be inspected in **Table 1**.

External data preprocessing

The external validation datasets were subjected to several preprocessing steps to match the MAGNet data structure and distribution. First, the samples were converted from raw counts or reads per kilobase million to CPM. These values were then log-transformed in such a way that values equal to 0 were set to the value of $\log_2(0.25)$. Extremely low log-transformed values were set to a threshold. The gene lists were also matched to the MAGNet dataset; extra genes were deleted and missing genes were left empty ("NA" values), displayed using Ensembl gene identifiers. Finally, the counts were rescaled to the range of values in the MAGNet dataset and empty values were set to 0.

Exploratory data analysis

A Principal Component Analysis (PCA) was performed on all datasets using the R package *PCAMethods*[23] and visualized with *ggplot2*[24].

ML algorithms

Two different ML strategies were applied in this study: binary and multi-class classifiers. First, we created binary models to distinguish between NF and DCM, NF and HCM, and NF and PPCM. Then, we developed multi-class algorithms to classify NF, DCM, HCM, and PPCM using the One-vs-Rest method[25], which consists of training one binary classifier per class (i.e., each class is fitted against all the other classes).

The input data for the ML models was the CPM gene expression from the MAGNet dataset, and the output was the label for the disease status of samples. An 80:20 ratio was chosen to split the data into training and test datasets.

In the first stage, the Lazy Predict Python library[26], which builds around twenty-five different classifiers, was used with default parameters. According to their performance metrics on the test dataset, the top 3 ensemble-based algorithms were selected, re-trained using various Python libraries (scikit-learn[27], XGBoost[28], and LightGBM[29]), and re-evaluated based on their performance on the test and external datasets. A detailed description of each performance metric can be consulted in **Table SI1**.

The ensemble-based models were chosen because of their interpretability and ability to extract the feature (i.e., genes) importance. In general, a feature with a higher importance index contributes the most to the prediction capacity of a ML model. After training the top 3 ensemble-based algorithms for each classification task, we obtained the top 500 most important features per algorithm and created consensus lists of genes shared among at least two models. To evaluate the prediction capacity of the consensus lists we trained new ML classifiers with these genes as the only features.

Biological interpretation

Based on two different studies by Japp. et al.[30] and Hershberger & Siegfried[31], a consensus list was created with the genes found to be relevant in DCM. This list was compared with the relevant features extracted from the NF/DCM binary classifier.

GOrilla[32] and Enrichr[33] were used for the GSE analysis. In the GOrilla platform, *"Homo sapiens"* was chosen as the organism. The "Two unranked lists of genes'' function was selected, which required two lists of genes, the target set (the consensus list of genes of the ML models), and the background set (a list of all the genes from the MAGNet dataset). The Enrichr platform only required the list of genes of interest as the input.

Model deployment

The deployment of the best binary classifier for DCM was done with the Streamlit Python library[34].

Ethical statement

The data from all studies (MAGNet and the 3 external datasets) were collected with the informed consent of the subjects or next of kin, and were approved by the relevant institutional ethics committees[16,18,20,22].

Results

Three external datasets were found that fit the search criteria. A summary of relevant features found in the datasets can be found in **Table 1**.

Table 1. Summary of the datasets for training and evaluating the ML models

Dataset (GEO ID)	Sequencing platform	Sample size	Data format
GSE141910 (MAGNet)	Illumina HiSeq 2500	166 NF, 166 DCM, 28 HCM, 6 PPCM	CPM
GSE46224	Illumina HiSeq 2000	8 NF, 8 DCM	RPKM
GSE55296	AB 5500xl Genetic Analyzer	10 NF, 13 DCM	raw counts
GSE116250	Illumina HiSeq 2500	14 NF, 37 DCM	RPKM

GEO = Gene Expression Omnibus; MAGNet = Myocardial applied genomics network; RPKM = Reads per kilobase million

Performance of ML algorithms

Using the test dataset to evaluate the performance of the ML algorithms, the best-performing classification task was NF/DCM, with perfect scores in the first five classifiers and high values (>0.85) in all performance metrics for the next 18 classifiers (**Table SI2**). Linear regression-based algorithms had the highest scores for all metrics, followed by ensemble-based algorithms, the latter which can be inspected in **Table 2** and **Figure 1A**. For the NF/HCM classification task, the algorithms were less effective, with only one model scoring perfectly and 11 scoring high (>0.85) across all metrics (**Table SI3**). The ensemble-based models still had high scores (**Figure SI1A**), except for the recall score for the XGBoost classifier, which can be observed in **Table 3**. The scores that decreased the most were the MCC and recall (**Table 3**). The NF/PPCM classification scored the poorest with a severe decrease in balanced accuracy, MCC, and precision (**Table SI4**). Precision, f1 score, ROC-AUC, and MCC were low for the ensemble-based-based models of the NF/PPCM classification task, as can be observed in **Table 4** and **Figure SI1B**.

Table 2. Performance metrics of the top 3 ensemble-based classifiers employed in the binary classification task of *NF/DCM* using the test data

Performance metric/Algorithm	RF	LGBM	XGBoost
Accuracy	0.99	0.99	0.97
Balanced Accuracy	0.99	0.99	0.97
Precision	0.97	0.97	0.94
Recall	1.00	1.00	1.00
F1score	0.99	0.99	0.97
MCC	0.97	0.97	0.94

RF = Random Forest; LGBM = Light gradient-boosting machine; XGBoost = Extreme Gradient Boosting; MCC = Matthews correlation coefficient

Table 3. Performance metrics of the top 3 ensemble-based classifiers employed in the binary classification task of NF/HCM using the test data

Performance	Decision	LGBM	XGBoost

metric/Algorithm	Tree		
Accuracy	0.97	0.97	0.95
Balanced accuracy	0.92	0.92	0.83
Precision	1.00	1.00	1.00
Recall	0.83	0.83	0.67
F1score	0.91	0.91	0.80
MCC	0.90	0.90	0.79

LGBM = Light gradient-boosting machine; XGBoost = Extreme Gradient Boosting; MCC = Matthews correlation coefficient

Table 4. Performance metrics of the top 3 ensemble-based classifiers employed in the binary classification task of NF/PPCM using the test data

Performance metric/Algorithm	LGBM	AdaBoost	DecisionTree
Accuracy	1.00	0.97	0.94
Balanced accuracy	1.00	0.99	0.97
Precision	1.00	0.50	0.33
Recall	1.00	1.00	1.00
F1score	1.00	0.67	0.50
MCC	1.00	0.70	0.56

LGBM = Light gradient-boosting machine; AdaBoost = Adaptive Boosting; MCC = Matthews correlation coefficient

The external data was used to validate the binary classification task of NF/DCM, as the external data only had DCM and NF samples. In the top 3 ensemble-based models, the only moderate scores (>0.75) were the recall for Random Forest and XGBoost and the f1 score for Random Forest (**Table 5**). The ROC curves (**Figure 1A**) also illustrated that the models evaluated on the external datasets were close to the random classifier, compared to the model evaluation on the test dataset, whose curves were close to the upper left corner (high performance).

Table 5. Performance metrics of the top 3 ensemble-based classifiers employed in the binary classification task of NF/DCM using the external data

Performance metric/Algorithm	RF	LGBM	XGBoost
Accuracy	0.66	0.52	0.62
Balanced accuracy	0.58	0.48	0.55
Precision	0.69	0.63	0.68
Recall	0.84	0.62	0.79
F1score	0.76	0.63	0.73
MCC	0.18	-0.04	0.12

RF = Random Forest; LGBM = Light gradient-boosting machine; XGBoost = Extreme Gradient Boosting; MCC = Matthews correlation coefficient

Model performance metrics for the multiclass models were high (>0.85) or moderate (>0.75), except for the balanced accuracy which was close to random performance (~0.5), as can be seen in **Table 6**. **Figure 1B** depicts the ROC curves for each etiology individually, where the NF and DCM curves approached the upper left corner (high performance), whereas the HCM and PPCM curves were closer to the middle line, depicting the performance of a random classifier. When tested on the external data, the models performed poorly (**Table SI5**), keeping consistent with the observations for the binary classification of NF/DCM.

Table 6. Performance metrics of the ensemble-based-based classifiers employed in the multiclass classification task using the test data.

Performance metric/Algorithm	RF	LGBM	XGBoost
Accuracy	0.91	0.91	0.89
Balanced accuracy	0.50	0.53	0.49
Precision	0.83	0.91	0.82
Recall	0.91	0.91	0.89
F1score	0.86	0.88	0.85
MCC	0.84	0.84	0.82

RF = Random Forest; LGBM = Light gradient-boosting machine; XGBoost = Extreme Gradient Boosting; MCC = Matthews correlation coefficient

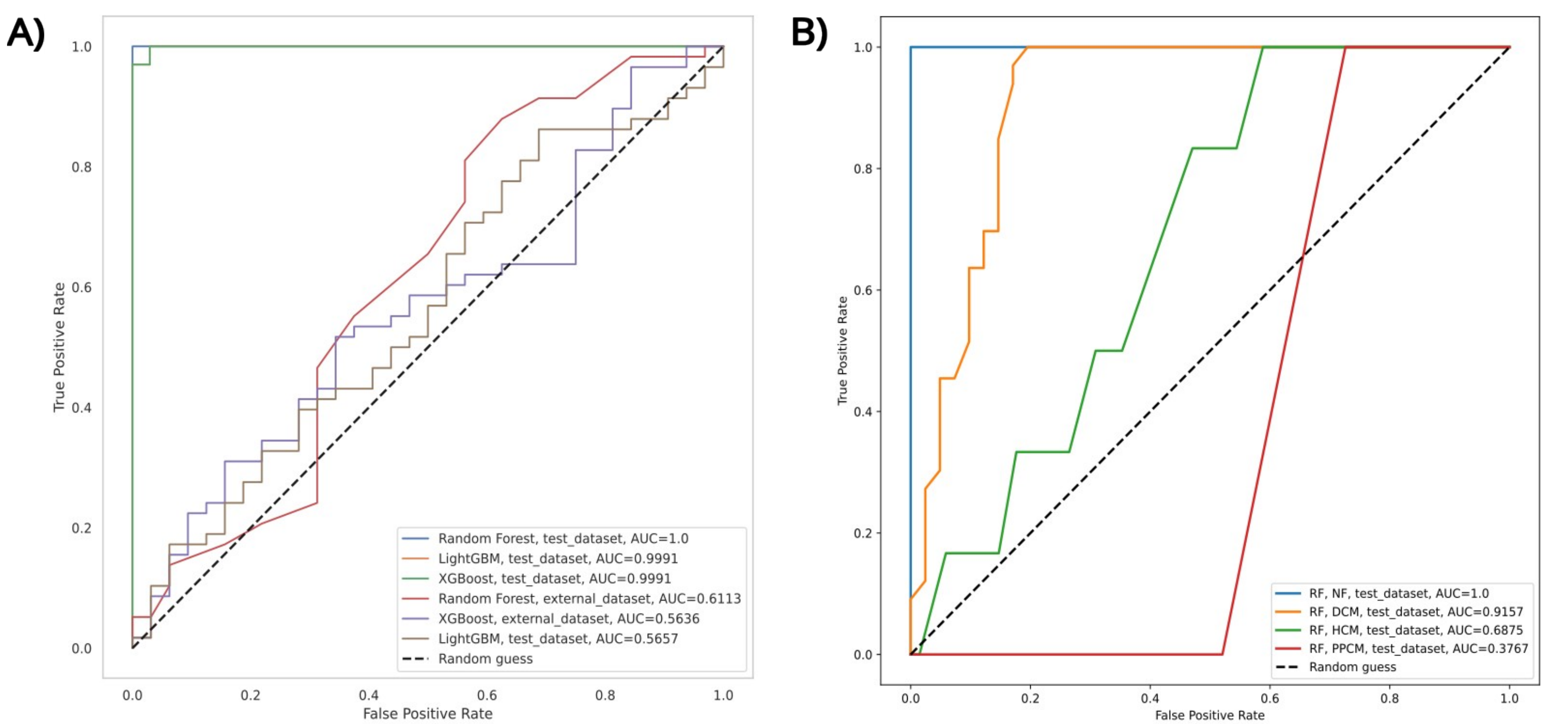


Figure 1. ROC curves of the top 3 ensemble-based-based models in A) the binary classification task of *NF/DCM* using test and external data and B) individual performances for each etiology in the multiclass classification task on test data. The ROC curves depict the balance between the false positives and the true positive rates. The upper left corner represents a perfect classifier, whereas the dashed line in the middle represents a random classifier. The area under the curve (AUC) is a numeric representation of the performance of the model.

Biological interpretation

Using the 500 most important genes in each of the top 3 ensemble-based models (Random Forest, LGBM, and XGBoost) from the NF/DCM classification task, 94 genes were found to be shared in at least 2 of the 3 models and therefore deemed relevant for DCM. **Figure 2A** shows a Venn diagram that summarizes the genes shared between all the models. After retraining the model with only these 94 genes as features, the performance of all models remains unchanged. In addition, we performed a similar analysis for the other classification tasks (**Figure SI2**).

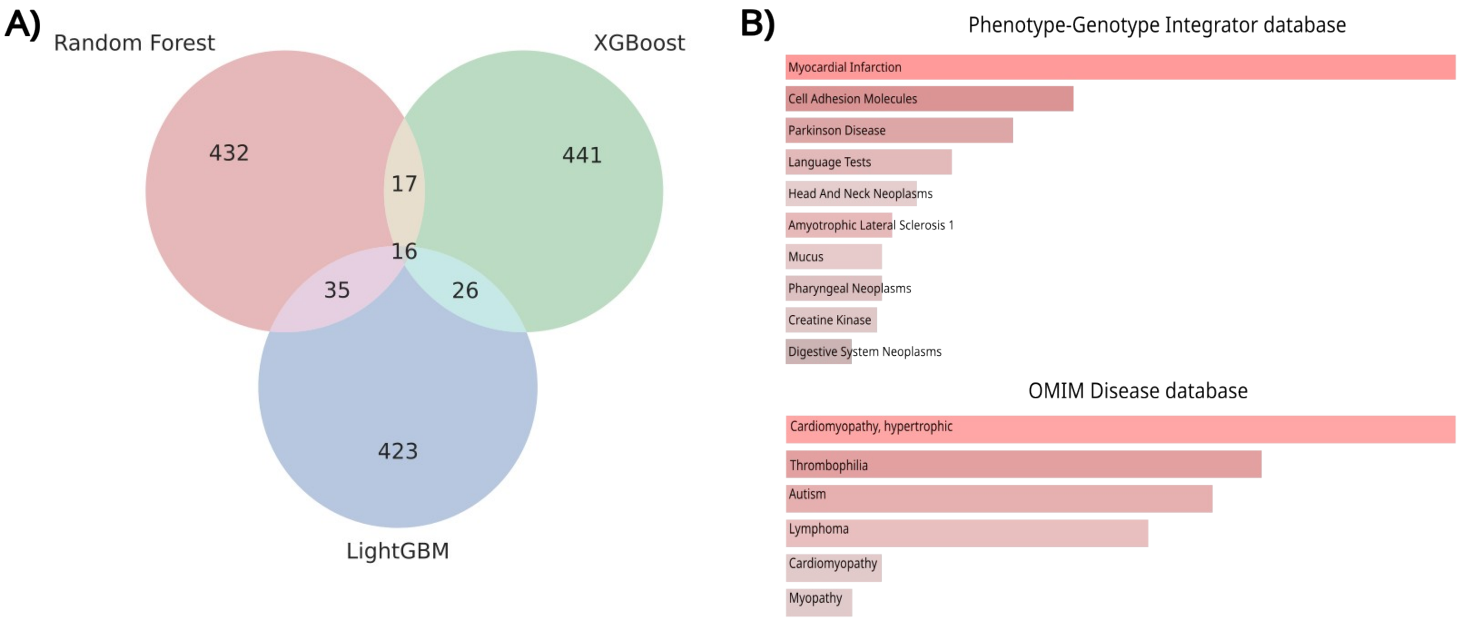


Figure 2. A) Venn diagram of the top 500 most important genes according to the best ensemble-based models for DCM from the binary classification task. B) Gene set enrichment results associated with heart diseases and cardiomyopathies obtained with the Enrichr web server[33].

Literature comparison and GSE analysis

When comparing the consensus gene list to the 2 sets from literature[30,31], only a single gene out of 94 was encountered, namely myosin heavy chain (MYH6), which has a role in cardiac muscle contraction. Mutations in this gene have been associated with cardiomyopathies, ventricular dysfunction, atrial septal defect, and sudden cardiac death[35–37].

Furthermore, the GSE analysis with Enrichr with the consensus gene list from the DCM/NF classification task identified some genes enriched in the myocardial infarction and HCM ontologies from the Phenotype-Genotype Integrator and OMIM Disease databases (**Figure 2B**). It is worth mentioning that only 2 databases out of 35 showed enriched gene sets with terms related to heart diseases and cardiomyopathies. Additionally, the same gene list yielded no results related to heart diseases using GOrilla, which identified ontology terms associated with response to lipid hydroperoxide, neural crest differentiation, endochondral bone growth, and cellular response to acid pH.

Discussion

ML performance

In all binary classification tasks, the performance of the ML algorithms was very good (>0.85), with the NF/DCM classification having near-perfect performance. This is in line with results from other authors that performed similar procedures[12]. The lower performance of NF/HCM and NF/PPCM classification tasks is understandable since the sample size of those etiologies was substantially lower, making the predictions more unreliable (**Tables 3-4, Figure SI1**). The disparity in performance and sample size is highlighted in the multiclass classifiers, where the surface-level acceptable results cover up the extremely poor performance of HCM and PPCM classification, underscored by the balanced accuracy (**Table 6, Figure 1B**). There is evidence that more training data improves the prediction of ML classifiers[13]. Nonetheless, no external datasets of PPCM were found in our search for external databases and only one result of a dataset with HCM was found, which itself was performed using microarray technology instead of RNA-seq (E-GEOD-1145) [38]. The lack of PPCM data is unsurprising as some classifications consider PPCM to be a subtype of DCM[3], however, there is evidence that HCM has a genetic component[4], so the lack of data for this etiology was unfortunate.

There is a clear disparity between the ability of the ML algorithms to classify data from the test dataset (that has the same data structure and distribution as the training set) and external data. While the performance of the binary ML algorithms on the test dataset was very good, the classification performed on external validation datasets was almost random (**Table 5, Figure 1A**). One explanation of the poor external validation can be related to the processing steps made for the external datasets.

The origin of the data in the MAGNet dataset (which stands at the basis of this study) is poorly reported. Information about tissue handling, normalization, filtering of lowly expressed genes, or other decisions is not available or extremely hard to find. These shortcomings prompted us to focus on altering the external data to be in the same unit and range as the MAGNet data for the comparison to be feasible.

When converting the external data into log-transformed CPM, a lot of the data had very small values (< -50) which effectively meant no gene was detected (expression = 0). Additionally, the gene lists of external datasets were matched to the gene list of the MAGNet dataset, so genes not originally present in the external datasets do not have assigned values. Because the ML algorithm cannot handle missing data, these unassigned values were set to 0. The high amount of zeroes (**Figure SI3**) skews the data distribution and can make prediction more difficult for the algorithm.

Moreover, scaling could have skewed the data even further. Before scaling, the external datasets showed a similar separation of data points between healthy and DCM in the first few principal components (**Figures SI4-7**). However, once the data was scaled, the principal component analysis plots showed abnormal data structures (**Figures SI8-10**). Therefore, although feature normalization has been shown to improve performance in ML algorithms[39], we must emphasize that any manipulation of the data can skew the results by severely altering the underlying data structure. This amount of data manipulation could make replicability in future research and applicability in clinical settings much more difficult. While processing the datasets is necessary to make the external datasets comparable to MAGNet dataset, it is clear that the current data processing procedures need to be rethought. Some alternative strategies include normalization and batch correction[10,40,41].

Remarkably, despite the fact that external validation is often cited as “the gold standard” for the validation data-driven approaches (including ML), to our knowledge, no study using RNA-seq data is attempting this method. If multiple datasets are available, it seems that the common practice is to include a mix of all datasets into the training set and leave a mix of data for the test data[12]. We suspect that this might allow the models to distinguish patterns based primarily on the expression information itself.

Biological interpretation

While we have found that classification and feature extraction are not difficult tasks for the ML algorithms, it's relevant to remark that traditional statistical methods like PCA and DGEA can also perform these feats. However, ML methods can still provide additional utility, especially for the latter task. DGEA handles genes independently from each other and yields a list of thousands of significant genes which can be difficult to interpret and rank in order of contribution. ML algorithms, on the other hand, take into account correlations and relationships between genes and can sort the features in order of importance. These two improvements allow for the recognition of more complex expression patterns and for the identification of potential genetic biomarkers for future study[13].

However, consensus gene selection by choosing features present in 2 out of 3 models is omitting a large amount of potentially relevant genes which are present only once in any of the individual models. This is problematic because if there were three important, perfectly correlated genes each ensemble model could, by chance, select one of them. None of these three hypothetical genes of interest would show in the consensus list, but they are still important. This concern is still valid when features are only highly correlated (as genes often are) and is exacerbated when there are more than just three correlated features.

Future Outlook

Due to the limitations discussed in previous sections, we suggest some improvements and developments. The most direct improvement is to employ better post-processing techniques or different ML strategies (e.g., cross-validation for splitting the data), in the hopes that either of these can improve external validation. An alternative approach could involve training a new algorithm using multiple datasets from different experiments so that the algorithm does not "overfit" the patterns of one particular dataset. Outside of this method, more datasets studying HCM and PPCM along with a more standardized sample and data processing procedures could improve the applicability of ML algorithms across datasets.

While the ML algorithms trained in this study did not perform as well as expected in the external data, the application of ML technology to omics data shows promise in several areas. First, the binary classifiers could be employed to give more objective and data-driven diagnoses. Second, feature selection shows potential for exploratory studies into genes of interest, biomarkers, or targets for treatments. Third, the web application improves data and model accessibility.

Data Availability

All of the code and data for this study can be found in our GitHub repository. Additionally, we launched a web application in which users can upload a single-patient CSV file with the expression of the 94 consensus genes. The web tool will then predict whether the sample belongs to a NF or DCM sample.

Supplementary Information

The supplementary tables and figures are available at Zenodo.

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